Quiz 2

CSE428 : Image Processing
Total Marks: 30
Time: 30 minutes

BRAC UNIVERSITY

Dept. of Computer Science and Engineering

Student ID:		**Section:**	
Name:			

1. **[2+4+4][CO1]**

A. **Compare correlation and convolution operations.** Under what condition are they equivalent?

B. **Explain the Prewitt edge detection operator.** Write the kernels for both horizontal and vertical edge detection.

C. **Output dimensions:** Given an input image of size 100×100 and a kernel of size 5×5, what is the output size for:

(i) Valid convolution (no padding)?
(ii) Same convolution (with appropriate padding)?

2. **[7][CO2]**

Design a 3×3 kernel that detects: **Diagonal edges at 45° angle (any of the diagonal edges)**. Show your kernels and explain why they work.

3. **[2+8+3][CO2]**

Given a 3*3 Gaussian kernel :

1	2	1
2	4	2
1	2	1

And the following noisy image (8 bit):

10	12	50
11	48	14
12	13	51

A. What will be the **appropriate pad size** for this kernel and image to keep the input and output dimension same?

B. Apply **convolution at the (0,0) pixel (pixel with intensity 10)**. Show the **kernel flipping**, padding, and every calculation step clearly. (use nearest padding according to appropriate pad size).

C. Explain why the result effectively reduces noise.